$$x = 10$$

Therefore, he can buy 10 boxes for the same amount.

Direct and indirect proportion is very much useful in project scheduling. A project can be any work that is time bound like building a house, construction of bridges, etc.





Exercise 4.2

1. Fill in the blanks.
   (i) 16 taps can fill a petrol tank in 18 minutes. The time taken for 9 taps to fill the same tank will be _____ minutes.
   (ii) If 40 workers can do a project work in 8 days, then_____ workers can do it in 4 days.
2. 6 pumps are required to fill a water sump in 1 hr 30 minutes. What will be the time taken to fill the sump if one pump is switched off?
3. A farmer has enough food for 144 ducks for 28 days. If he sells 32 ducks, how long will the food last?
4. It takes 60 days for 10 machines to dig a hole. Assuming that all machines work at the same speed, how long will it take 30 machines to dig the same hole?
5. Forty students stay in a hostel. They had food stock for 30 days. If the students are doubled then for how many days the stock will last?
6. Meena had enough money to send 8 parcels each weighing 500 grams through a courier service. What would be the weight of each parcel, if she has to send 40 parcels for the same money?
7. It takes 120 minutes to weed a garden with 6 gardeners If the same work is to be done in 30 minutes, how many more gardeners are needed?
8. Neelaveni goes by bi-cycle to her school every day. Her average speed is 12km/hr and she reaches school in 20 minutes. What is the increase in speed, if she reaches the school in 15 minutes?
9. A toy company requires 36 machines to produce car toys in 54 days. How many machines would be required to produce the same number of car toys in 81 days?

Objective type questions

10. 12 cows can graze a field for 10 days. 20 cows can graze the same field for_____ days.
    (i) 15 (ii) 18 (iii) 6 (iv) 8
11. 4 typists are employed to complete a work in 12 days. If two more typists are added, they will finish the same work in _____ days.
    (i) 7 (ii) 8 (iii) 9 (iv) 10

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